AttentionDTA: Drug–Target Binding Affinity Prediction by Sequence-Based Deep Learning With Attention Mechanism

Qichang Zhao, Guihua Duan, Mengyun Yang, Zhongjian Cheng, Yaohang Li, and Jianxin Wang

Abstract—The identification of drug–target relations (DTRs) is substantial in drug development. A large number of methods treat DTRs as drug-target interactions (DTIs), a binary classification problem. The main drawback of these methods are the lack of reliable negative samples and the absence of many important aspects of DTR, including their dose dependence and quantitative affinities. With increasing number of publications of drug–protein binding affinity data recently, DTRs prediction can be viewed as a regression problem of drug–target affinities (DTAs) which reflects how tightly the drug binds to the target and can present more detailed and specific information than DTIs. The growth of affinity data enables the use of deep learning architectures, which have been shown to be among the state-of-the-art methods in binding affinity prediction. Although relatively effective, due to the black-box nature of deep learning, these models are less biologically interpretable. In this study, we proposed a deep learning-based model, named AttentionDTA, which uses attention mechanism to predict DTAs. Different from the models using 3D structures of drug–target complexes or graph representation of drugs and proteins, the novelty of our work is to use attention mechanism to focus on key subsequences which are important in drug and protein sequences when predicting its affinity. We use two separate one-dimensional Convolution Neural Networks (1D-CNNs) to extract the semantic information of drug's SMILES string and protein's amino acid sequence. Furthermore, a two-side multi-head attention mechanism is developed and embedded to our model to explore the relationship between drug features and protein features. We evaluate our model on three established DTA benchmark datasets, Davis, Metz, and KIBA. AttentionDTA outperforms the state-of-the-art deep learning methods under different evaluation metrics. The results show that the attention-based model can effectively extract protein features related to drug information and drug features related to protein information to better predict drug target affinities. It is worth mentioning that we test our model on IC50 dataset, which provides the binding sites between drugs and proteins, to evaluate the ability of our model to locate binding sites. Finally, we visualize the attention weight to demonstrate the biological significance of the model. The source code of AttentionDTA can be downloaded from https://github.com/zhaoqichang/AttentionDTA_TCBB.

Index Terms—Virtual screening, drug–target affinity, attention mechanism, deep learning

1 Introduction

The newly discovered drug–target relations (DTRs) are critical in the drug discovery and development. At present, many compounds have not been used as drugs. According to statistics, there are approximately 13,580 drug entries stored in the DrugBank database, including 2,635 approved small molecule drugs, 1,378 approved biological agents and 6,373 experimental drugs. Of the approximately 20,000 human proteins, only 3,150 are related to these drugs while the estimated number of human drug targets is approximately 4,500, which corresponds to about 22% of the human genome [1]. Therefore, a large number of potential DTRs need to be verified. The detection of these unknown relations between compounds and disease-related proteins is essential for drug development and drug repositioning [2], [3], [4].

A common method for DTR prediction is *in vitro* screening experiments which are costly, labor-intensive, and time-consuming with high failure rates [5]. Moreover, it is not a realistic way to verify every possible drug–target pair considering the enormous chemical and proteomic spaces. This can be accelerated by introducing virtual screening (VS) [6]. Therefore, predicting the DTRs through computational methods has attracted increasing attention [7], [8]. It is reported that deep learning-based methods have shown more powerful ability for DTR prediction compared with conventional machine learning-based methods [9]. This is because of the capacity of deriving the information hidden in the data by hierarchically learning complex and non-linear features from raw input data [10], [11], [12]. With a large amount of biological activity data published in recent years,

- *Qichang Zhao, Guihua Duan, Zhongjian Cheng, and Jianxin Wang are with the School of Computer Science and Engineering, Central South University, Changsha, Hunan 410083, China, and also with Hunan Provincial Key Lab on Bioinformatics, Central South University, Changsha, Hunan 410083, China. E-mail: {137241304, zj_cheng}@qq.com, duangh@csu.edu.cn, jxwang@mail.csu.edu.cn.*
- *Mengyun Yang is with the School of Science, Shaoyang University, Shaoyang, Hunan 422000, China. E-mail: mengyun_yang@126.com.*
- *Yaohang Li is with the Department of Computer Science, Old Dominion University, Norfolk 23529 USA. E-mail: yaohang@cs.odu.edu.*

Manuscript received 17 August 2020; revised 2 February 2022; accepted 22 April 2022. Date of publication 26 April 2022; date of current version 3 April 2023.
This work was supported in part by the National Natural Science Foundation of China under Grants 62072473, and 61972423, and in part by 111 Project under Grant B18059. This work was also supported in part using computing resources at the High Performance Computing Center of Central South University.
(Corresponding authors: Guihua Duan and Jianxin Wang.)
Digital Object Identifier no. 10.1109/TCBB.2022.3170365

deep learning-based approaches make it possible to predict relations between drugs and target proteins on a large scale.

Most of the deep learning-based methods have treated the prediction of DTRs as drug–target interactions (DTIs), a binary classification, to determine whether a drug–target pair interacts or not. Different deep learning models have been developed to predict DTIs and get good performance including restricted Boltzmann machines [13], [14], deep neural networks [15], [16], stacked auto-encoders [17], [18], and deep belief networks [19], [20]. Du *et al.* [21] proposed a wide and deep learning framework in which the wide model is a generalized linear model and the deep model is a deep feed-forward neural network to predict interactions. Zheng *et al.* [22] proposed a hybrid Neural Network, named DTI-RCNN, that integrated recurrent neural network (RNN) with convolutional neural network (CNN). Lee *et al.* [23] proposed a model with amino acid sequence and drug fingerprint string as input, named DeepConv-DTI. DeepConv-DTI used a variety of 1D-CNNs with convolution kernels of different sizes to capture the local residual conformation patterns of the protein, effectively reducing the risk of improper convolution kernel size. Tsubaki *et al.* [24] developed a new DTI prediction approach by combining a graph neural network (GNN) for drugs, a CNN for proteins and an attention mechanism [25] to integrate the representations. Cheng *et al.* [26] proposed an end-to-end deep learning method to predict DTIs based on the graph attention network and multi-head self-attention mechanism. Rifaioglu *et al.* [27] proposed a large-scale DTI prediction system, named DEEPScreen. DEEPScreen takes compound image as input and constructs a drug image classification sub-network based on deep convolutional neural network with reference to the classic image classification network for each target protein to accurately predict interacting small molecule ligands for a unique target protein. Researchers have also started to draw on some efficient models from other fields to deal with DTI prediction. Zhang *et al.* [28] proposed a novel Visual Question Answering (VQA) mode to predict DTIs, by representing proteins with 2D distance map (Image), and drugs with molecular linear notation (String). Inspired by Transformer's [29] powerful ability to capture features between two sequences, Chen *et al.* [30] proposed a novel Transformer neural network, named TransformerCPI, to predict DTIs. TransformerCPI used a gated convolutional network [31] with 1D-CNN and gated linear unit to replace the original self-attention encoder in Transformer to better deal with drug and protein sequences.

An inevitable problem in DTI prediction is the absence of reliable negative instances (confident non-interactions) [32]. Randomly selecting negative cases from the set of unlabeled DT pairs cannot always guarantee that the model can distinguish positive samples from unknown samples very well. Because of the use of inappropriate datasets, many DTI prediction methods are pointed out to make predictions according to the ligand patterns rather than drug–target interaction features, resulting in a mismatch between theoretical modeling and practical application [30]. Besides, in reality, the DTR is experimentally measured in a continuous scale that quantifies how strongly a drug binds to a target. Biochemical interactions originate from the interaction between drug's atoms and amino acids on receptor proteins. The binding affinity is affected by non-covalent molecular interactions, such as van der Waals forces, hydrophobic forces, hydrogen bonding, and electrostatic interactions. The advantages of the prediction of drug–target affinities are to avoid the influence of negative sample selection and the hidden ligand bias in the datasets and provide more practical and useful information [9], [32]. Recently, deep learning has made good progress in the prediction of DTAs which deal with 3D structures directly. AtomNet [33] was the first deep learning model to process the 3D structure of the drug–target complex by three-dimensional convolutional neural networks (3D-CNNs). Gomes *et al.* [34] proposed the Atomic Convolutional Neural Network (ACNN) for predicting binding affinities. ACNN contains two special operations, atom type convolution and radial pooling, which are used to extract features encoding local chemical environments and reduce computational complexity, respectively. Inspired by SqueezeNet [35], Jimenez *et al.* [36] also proposed an end-to-end framework based on 3D-CNNs for predicting drug–target absolute affinities. In order to reduce the computational complexity and achieve excellent prediction performance, Li *et al.* [37] proposed an effective and light-weight neural network based on 3D-CNNs, named DeepAtom, to automatically extract binding related atomic interaction patterns. Nevertheless, these deep learning methods rely on the actual 3D structure of drug–target complex but most drug–target pairs of the binding sites do not have experimentally solved co-crystal structures. In addition, the 3D grid representation of the drug–target complex is very sparse and computationally costly. The broad use of 3D-based approaches at a large scale has been limited. Sequence- and graph-based methods can effectively avoid the lack of 3D structures and computational cost problems. DeepDTA [38] is a novel sequence-based model using the drug SMILES string and the amino acid sequence of the protein as inputs and consists of two independent 1D-CNN blocks that effectively predict DTAs. WideDTA [39] added two extra features, ligand max common structures (LMCS) and protein motifs and domains (PDM), on the basis of DeepDTA to improve model performance. GraphDTA [40] takes the graph structure of drug as input and uses the excellent graph neural networks (GNNs), i.e., graph convolutional networks (GCNs) [41], graph attention network (GAT) [42] and Graph Isomorphism Network (GIN) [43], to replace the 1D-CNN and achieved good performance. Although the aforementioned deep learning methods have demonstrated certain success, their black-box characteristic limits their interpretability in biological perspectives.

To tackle this problem, we propose an end-to-end model, named AttentionDTA, to predict the binding affinities of drug–protein pairs with attention model. The present paper extends a previously published conference paper [44]. The previous conference paper proposes two attention calculation modes, Co-attention and Cross-attention, and presents a comparative experiment with DeepDTA under the setting S1. In the present paper we have extended this work. First, we develop a novel two-side multi-head attention mechanism and design the ablation study to explore the influence of attention calculation mode on model performance. Second, we redesign the experimental comparison program.

TABLE 1
Summary of the Datasets

Datasets	Proteins			Drugs			Affinities
	number	Maximum length	Average length	number	Maximum length	Average length	
Davis	442	2,549	788	68	103	64	30,056
Metz	170	2,527	731	1423	112	44	35,259
KIBA	229	4,128	728	2,111	590	58	118,254

Our proposed model AttentionDTA is performed on three benchmark datasets, the Davis [45], the Metz [46] and the KIBA [47] datasets, and is compared with two state-of-the-art approaches, i.e., DeepDTA and GraphDTA. We set four different data division settings to get a comprehensive comparison. Third, to evaluate the ability of our model to locate binding sites, we test our model on IC50 dataset [48], which provides the binding sites between drugs and proteins. Moreover, to better display the visualization, we add the new visualization by mapping the regions with the attention weight onto the protein sequence.

2 Materials and Methods

2.1 Datasets

We evaluated our model on three different Kinase datasets, the Davis [45], the Metz [46] and the KIBA [47] datasets, which are widely used as benchmarks for DTA prediction [32], [38], [40], [49]. The binding affinity values are measured in dissociation constant (K_d), log transformed kinase inhibition constant (pK_i), and KIBA metric [47], respectively. The KIBA score can be viewed as a model-based summary statistic of all the available information captured by the half maximal inhibitory concentration ($IC50$), inhibition constant (K_i), and K_d, where the consistency between them is optimized to make use of the complementary information. The K_d values in Davis dataset are usually transformed into logspace (pK_d) to reduce the large variance:

$$pK_d = -\log_{10}\left(\frac{K_d}{1e9}\right)), \tag{1}$$

which is ranging from 5.0 to 10.8. The pK_i of Metz range from 0.0 to 12.0, and the KIBA score range from 0.0 to 17.2. Table 1 presents the details of three datasets.

2.2 AttentionDTA Model

AttentionDTA's design inspiration comes from the concept that different drugs have different binding sites on the same protein, and vice versa. As mentioned earlier, we adapt a popular deep learning architecture, 1D-CNN, to extract useful information from the sequence. 1D-CNN has the capability to capture important local patterns from the entire space. With the convolution filters sliding on proteins or SMILES strings, the different combinations of amino acids or substructures of the drug are captured. To model non-covalent interactions between atoms and amino acids, we incorporate a two-side multi-head attention mechanism in our model. We calculate the attention scores between drug's and protein's subsequences in this module. The vector representations of subsequences are strengthened or weakened according to the attention scores. Attention mechanism incorporates the notion of relevance by allowing the model to dynamically pay attention to certain parts of the drug and target that help in predicting DTAs effectively, but inevitably, this introduces an additional consumption of computational resources. After that, the drug representation capturing relevant information of protein and the protein representation capturing relevant information of drug are obtained. An workflow of the AttentionDTA model is shown in Fig. 1.

2.2.1 Input Representation

As a typical sequence-based model, the inputs of AttentionDTA are the amino acid sequences of the proteins and SMILES string of the drugs. We use integer/label encoding that uses integers for the characters to encode the inputs. The SMILES strings are made up of 64 different characters. For protein sequences, there are 20 different categories. Here we construct two dictionaries for drug strings and protein sequences, such as {'C': 1, 'N': 2, 'O': 3, '=': 4, '(': 5, ')': 6 etc.} for drug. For example, a SMILES string, 'CC(C(=O)O) O,' can be represented as follows,

$$[\mathrm{C\ C\ (C\ (O)\ O)\ O\ }] = [1\ 1\ 5\ 1\ 5\ 4\ 3\ 6\ 3\ 6\ 3].$$

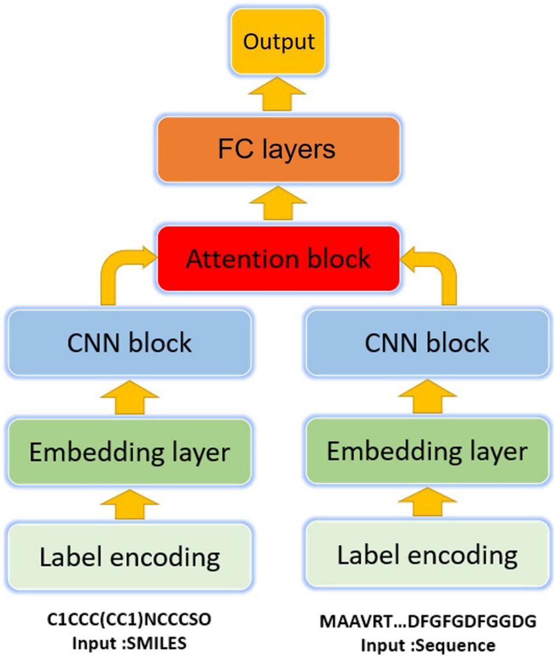


Fig. 1. AttentionDTA model with attention block to correlate drug and protein information.

After label encoding, we use two character embedding layers to transforme SMILES strings and protein sequences into embedded matrices. Hence, the output of the embedding layers are shown by drug-embedded matrix $D_e \in \mathbb{R}^{l_d \times F_e}$ and protein-embedded matrix $P_e \in \mathbb{R}^{l_p \times F_e}$, where l_d is the length of the SMILES sequence, l_p is the length of the protein sequence, and F_e is the embedding dimension.

2.2.2 1D-CNN Blocks

In our approach, two different 1D-CNN blocks are used: one for the protein-embedded matrix and one for the drug-embedded matrix.

Taking drug feature modeling as an example, given drug-embedded matrix $D_e = \{d_1, d_2, \ldots, d_{l_d}\} \in \mathbb{R}^{l_d \times l_e}$, a convolution operation can be expressed as

$$d_i^c = f(w \cdot d_{i:i+h-1} + b), \tag{2}$$

where h is the window size, $w \in \mathbb{R}^{h \times l_e}$ is the convolving kernel, $b \in \mathbb{R}^1$ is a bias term, $f(.)$ is a non-linear function such as the Rectified Linear Unit (ReLU), and $d_{i:i+h-1}$ is a subsequence which refers to the concatenation of characters d_i, d_{i+1},..., d_{i+h-1}.This operation is applied to each possible window of characters in the sequence $\{\mathrm{d}_{1:h}, \mathrm{d}_{2:h+1}, \ldots, \mathrm{d}_{l_d-h+1:l_d}\}$, from left to right, to generate a new feature representation,

$$D^c = \left[d_1^c, d_2^c, \ldots, d_{l_d-h+1}^c\right]. \tag{3}$$

Hence, after the padding operator, the output of 1D-CNN blocks are shown by drug matrix $D_c \in \mathbb{R}^{l_d \times l_c}$ and protein-embedded matrix $P_c \in \mathbb{R}^{l_p \times l_c}$, where l_c is the number of convolution kernels.

2.2.3 Attention Model

In this section, to model the non-covalent interactions between atoms and amino acids, we have introduced a two-side multi-head attention mechanism to our model. The inputs to our multi-head attention layer are the outputs of 1D-CNN blocks, drug matrix $D_c \in \mathbb{R}^{l_d \times l_c}$ and protein matrix $P_c \in \mathbb{R}^{l_p \times l_c}$.

In out model, the head number is set to K. For k-th head, to obtain sufficient expressive power to transform the input features into higher-level features and decouple attention mechanism from feature extraction, at least one learnable linear transformation is required. To that end, as an initial step, a shared linear transformation, parametrized by a weight matrix,$W^k \in \mathbb{R}^{l_c \times l_a}$, is applied to drug matrix D_c and protein matrix P_c:

$$D_a^k = \mathrm{ReLU}(D_c W^k + b), \tag{4}$$

$$P_a^k = \mathrm{ReLU}(P_c W^k + b). \tag{5}$$

We then compute the dot product-based scalar values α between D_a^k and P_a^k as the attention weights:

$$\alpha^k = \mathrm{Tanh}\left(D_d^k P_p^{kT}\right). \tag{6}$$

The final attention score is the average of K independent attention mechanisms:

$$\alpha = \frac{1}{K}\sum_{i=1}^{K} \alpha^i \tag{7}$$

The attention weights $\alpha \in \mathbb{R}^{l_d \times l_p}$, can be assumed to represent the interaction strength between the subsequences of drug and the subsequences of protein.

Next, the drug attention vector $\alpha_d \in \mathbb{R}^{l_d}$, which can be interpreted as attention scores of the expression of each position in the sequence of the drug to the whole protein, is calculated by applying row-wise sum operation to α. While protein attention vector $\alpha_p \in \mathbb{R}^{l_p}$ is calculated by applying column-wise sum operation to α. Each bit in the vector represents the extent to which the corresponding subsequence of protein is related to the drug representation.

$$\alpha_d = \sum_{j=1}^{M} \alpha_{i,j}. \tag{8}$$

$$\alpha_p = \sum_{i=1}^{N} \alpha_{i,j}. \tag{9}$$

We repeat both α_d and α_p for l_c times, so that $\alpha_d \in \mathbb{R}^{l_d \times l_c}$ and $\alpha_p \in \mathbb{R}^{l_p \times l_c}$ become matrices. We update drug's and protein's representations by α_d and α_p, respectively.

$$D_\alpha = \alpha_d \odot D_c, \tag{10}$$

$$P_\alpha = \alpha_p \odot P_c, \tag{11}$$

where $\odot$ denotes element-wise product.

We then apply a global-max pooling operation over D_α and P_α to obtain the maximum values. The idea is to capture the most important feature with the highest value for each feature map:

$$\mathbf{d}_{\text{drug}} = \mathrm{Globalmaxpooling}(D_\alpha), \tag{12}$$

$$\mathbf{p}_{\text{protein}} = \mathrm{Globalmaxpooling}(P_\alpha). \tag{13}$$

2.2.4 Output Block

The outputs of attention module, $\mathbf{d}_{\text{drug}}$ and $\mathbf{p}_{\text{protein}}$, are concatenated and feed into multi-layer perceptron (MLP). This MLP consists of 5 layers. The input layer receives outputs from attention model. We use 1,024 nodes for the next two fully connected (FC) layers, each followed by a dropout layer. Dropout [50] is adopted to prevent the deep learning model from overfitting. In the learning process, it reduces the dependence among nodes, thereby regularization of the neural network and reduces its structural risk by randomly returning the weights or outputs of hidden layers to zero. The third FC layer consists of 512 nodes and is followed by the output layer without dropout.

The activation function that we used in FC layers is Leaky Rectified Linear Unit (Leaky ReLU) [51]. The aim is to minimize the difference between the label and the prediction score during training. We use Mean Squared Error (MSE) as the loss function, in which $\hat{y}$ is the prediction value of model, y corresponds to the actual label, and n indicates the number of samples.

$$loss = \frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i)^2 \tag{14}$$

2.3 Implementation and Hyperparameter Search

2.3.1 Implementation

The training details of these neural networks are described as follows. Adam [52] is used as the optimization algorithm to train the networks with the default learning rate of 0.0001. We use embedding layer to represent characters with 128-dimensional dense vectors. The two CNN blocks consist of three 1D-convolutional layers with 32, 64, and 96 filters for both drugs and proteins. Because protein representation is longer than drug representation, the window sizes of at the three 1D-convolutional layers are 4, 6, 8 for drugs and 4, 6, 12 for proteins. We use 5-fold cross-validation with 10 repetitions under different random seeds to assess the predictive ability of our model. The algorithm is coded based on Pytorch [53]. We use a Linux server with 80 logical CPU cores and two Nvidia Geforce RTX 2080 Ti (12GB).

2.3.2 Hyperparameter Search

AttentionDTA has 4 important hyperparameters: the learning rate, the batch size, the number of attention heads, and the dropout rate. These hyperparameters are determined by grid-search on the validation MSE. In grid-search, the learning rate is in {0.1, 0.01, 0.001, 0.0005, 0.0001, 0.00001}; the batch size is in {32, 64, 128, 256, 512, 1024}; the number of attention heads is in {1, 2, 4, 8, 10, 16}; the dropout rate is in {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}. The optimized learning rate, batch size, the number of attention heads, and dropout rate are 0.0001, 128, 8, and 0.1, respectively.

3 Experiments and Results

In the following subsections, we compare the performance of these four attention modes on sequence information through experiments. We utilize the MSE, Mean Absolute Error (MAE) and Coefficient of Determination, denoted R^2, to measure the performance. We compare our model with K-nearest neighbor algorithm and the current state-of-art methods that we choose as the baselines, namely DeepDTA [38] and GraphDTA [40] on four experimental settings to make a comprehensive comparison. We provide more details about these baselines and the experimental setup, as well as our results in the following subsections.

3.1 Baselines

3.1.1 1-NN

1-nearest neighbor (1-NN) algorithm is a basic regression method. We refer to 1-NN based on the drug and protein similarity as 1-NN_drug and 1-NN_protein, respectively. As for 1-NN_drug, given the drug A and protein B, we rank the ligands of protein B in the training set according to their similarity to drug A. We choose the most similar drug A′ and set the affinity of drug A′ and protein B as the affinity value for drug A binding to protein B. The process of 1-NN_protein is similar with 1-NN_drug, but is based on protein similarity. The drug and protein similarities are generated by RDKit python package and CLUSTALW (online tool: www.genome.jp/tools-bin/clustalw), respectively.

3.1.2 Deepdta

DeepDTA [38] comprises two separate CNN blocks, each of which aims to learn representations from SMILES strings and protein sequences. For each CNN block, DeepDTA uses three consecutive 1D-convolutional layers with increasing numbers of filters. The number of filters in the second and third layers is twice and three times more than that in the first layer respectively. The convolutional layers are then followed by the maxpooling layer. The final features of the max-pooling layers are concatenated and feed into three FC layers. DeepDTA use 1,024 nodes in the first two FC layers, each followed by a dropout layer of rate 0.1. The third layer consists of 512 nodes and is followed by the output layer.

3.1.3 Graphdta

GraphDTA [40] is a model based on drug graph structure and amino acid sequence. The model uses different graph convolution models to extract the features of the drug graph structure, and uses 1D-CNN and the global maximum pooling layer to obtain the protein feature vector. The two parts of the feature are concatenated and feed into the fully connected network to obtain the prediction results. The paper provides four different graph convolution modules: GCN [41], GAT [42], GIN [43] and a combined model GAT_GCN. According to the experimental results provided in [40], the GIN model achieve the best experimental results. Therefore, we choose GIN as a representative as a representative GraphDTA model in the later experiment comparison.

DeepDTA has reported the best hyperparameters on the KIBA dataset by cross validation. GraphDTA has reported the hyperparameters they used. In order to ensure the effectiveness and fairness of the comparison, we follow the same hyperparameter setting described in the papers of baselines and use the open source code provided in DeepDTA and GraphDTA to conduct the experiment, and only modify the corresponding experiment settings and dataset division. Among them, DeepDTA is implemented with Keras [54] with Tensorflow [55] backend while GraphDTA is implemented with Pytorch [53] and PyTorch-Geometric [56].

3.2 Evaluation Metrics

We use the three most common evaluation metrics, including MSE, MAE, and R^2, in regression problems to evaluate the performance of our model and baselines (formulations are given below),

$$\mathrm{MSE}(y, \hat{y}) = \frac{1}{n_{\mathrm{samples}}} \sum_{i=1}^{n_{\mathrm{samples}}} (y_i - \hat{y}_i)^2, \tag{15}$$

$$\mathrm{MAE}(y, \hat{y}) = \frac{1}{n_{\mathrm{samples}}} \sum_{i=1}^{n_{\mathrm{samples}}} |y_i - \hat{y}_i|, \tag{16}$$

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^{n_{\mathrm{samples}}} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_{\mathrm{samples}}} (y_i - \bar{y})^2}, \tag{17}$$

where $\hat{y}_i$ is the predicted value of the i-th sample, y_i is the corresponding true value, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is the average of the true values, and n_{samples} is the number of samples.

TABLE 2
The Performance Under Different Truncation Settings on KIBA Dataset

The length of drug	The length of protein	Clipping ratio	MSE(std)	MAE(std)	R2(std)	Time	Memory
avg	avg	62.89%	0.1804(0.0054)	0.2388(0.0037)	0.7462(0.0042)	5.1 h	3,056MiB
80	1,000	23.17%	0.1562(0.0016)	0.2353(0.0038)	0.7528(0.0037)	5.5 h	5,512MiB
100	1,200	13.67%	0.1484(0.0066)	0.2128(0.0040)	0.7881(0.0055)	8.5 h	6,971MiB
max	1,200	12.89%	0.1489(0.0056)	0.2132(0.0038)	0.7875(0.0055)	9.2 h	11,032MiB
100	max	0.80%	0.1491(0.0077)	0.2141(0.0057)	0.7868(0.0039)	24.5 h	11,885MiB
max	max	0.00%	0.1486(0.0068)	0.2130(0.0051)	0.7879(0.0053)	35.6 h	17,445MiB

3.3 Experimental Settings

Let the training set for the model consist of a set S_{train} of drug–target pairs $s_{train} = (d_{train}, t_{train})$ and D_{train} and T_{train} denote the space of drugs and targets in the training set $S_{train} \subseteq D_{train} \times T_{train}$, respectively. Let the testing set for the model consist of a set S_{test} of drug–target pairs $s_{test} = (d_{test}, t_{test})$. In order to make a comprehensive comparison of various methods, four different cross-validation (CV) settings [32] are performed as follows:

- S1. Both d_{test} and t_{test} are encountered in the training set: $d_{test} \in D_{train}$ and $t_{test} \in T_{train}$.
- S2. The protein targets in the testing set appeared in the training set: $t_{test} \in T_{train}$, while the drugs in the testing set did not appear in the training set: $d_{test} \notin D_{train}$.
- S3. The drugs in the testing set appeared in the training set: $d_{test} \in D_{train}$, while the protein targets in the testing set did not appear in the training set: $t_{test} \notin T_{train}$.
- S4. Both d_{test} and t_{test} are not encountered in the training set: $d_{test} \notin D_{train}$ and $t_{test} \notin T_{train}$.

The setting S1 is the most widely used experimental setting in the virtual screening model. Under this condition, it is assumed that the original drug–target interaction matrix is sparse, and its purpose is to predict missing values without exceeding the space of known drugs and targets. The setting S2 and S3 are more compatible with practical applications. During the model training phase, only part of the drug or target information is available. In the most challenging setting S4, neither the drugs nor the proteins in the test phase were present in the training set. In chemogenomics-based DTA modeling, abstract features of drugs and targets are used to predict the relations between drugs and targets. Effective features to distinguish relations is a key component that the model should learn from the dataset. Therefore, models should have effective capabilities of extracting features to perform well in the setting S2, S3, and S4. Under the setting S3 and S4, we calculate the identity of the protein sequences in the training set and test set and delete proteins with sequence identity over 25% in the same set to ensure the independence of protein sequences between the training set and the test set.

We adapt a 5-fold cross-validation strategy to train and test our model and baselines in four settings. We divide the dataset into 5 parts on average according to different experimental settings and take one part as the test set and the remaining four parts as the training set. For example, under the setting S4, we randomly divide the drugs and proteins into 5 groups, and select a group from each to construct a drug–protein pair set to form a test set, and the remaining drug–protein pair set form the training set. These five models are trained with the same hyperparameters.

3.4 The Truncation of Inputs

The length of drug and protein sequences is variable. In order to improve the computational efficiency of our model, we set the maximum length of drugs and proteins. The sequences that are longer than the maximum length are truncated, whereas shorter sequences are 0-padded. We have designed an experiment to explore the changes of model performance under different truncation settings. We also recorded the clipping ratio (the percentage of samples that were truncated), time and memory used in training model. The performance on KIBA dataset are presented in Table 2. The "avg" means setting the maximum length of the strings to the average length and the "max" means using the whole strings without truncation. As can be seen from Table 2, when the setting length is larger than the average length of sequences, the performance of our model is robust to the setting length. The setting of 100 for drugs and 1,200 for proteins gets the best results. So, we set the length of the drug strings to 100 and the length of the proteins to 1,200. This truncation setting can effectively reduce model training time and computation cost without performance loss.

3.5 Comparison of AttentionDTA and Baselines Under Different Experimental Settings

In this subsection, we present the comparison results of our AttentionDTA and baselines under S1, S2, S3 and S4 CV settings. The best results in each row are in **bold** and the second best results are <u>underlined</u>.

Table 3 show the performance of each method on benchmark datasets under the setting S1. It can be seen that under the setting S1, AttentionDTA achieved the best results on both datasets. AttentionDTA leads to more significant improvement on the KIBA dataset than that on the Davis and Metz dataset, due to the fact that the KIBA dataset has a larger data volume. As expected in many deep learning algorithms, with increasing amount of data, AttentionDTA can better learn from the data to model the relationship between drugs and target features and extract more effective drug and protein features.

Table 4 presents the results under the setting S2. On the Davis and Metz datasets, the effects of DeepDTA and GraphDTA decrease significantly, and even the value of DeepDTA on the R^2 indicator is negative on Davis dataset,

TABLE 3
Performance Comparison of Our Model and Baselines on Benchmark Datasets Under the Setting S1

Datesets	Metrics	1-NN_drug	1-NN_protein	DeepDTA	GraphDTA	AttentionDTA
Davis	MSE(std)	1.0046(0.0117)	1.0057(0.0126)	0.2397(0.0046)	0.2623(0.0041)	**0.1741**(0.0070)
	MAE(std)	0.5351(0.0023)	0.5423(0.0025)	0.2541(0.0093)	0.3040(0.0042)	**0.2044**(0.0070)
	R^2(std)	-0.1756(0.0137)	-0.1802(0.0148)	0.6984(0.0057)	0.6931(0.0039)	**0.7409**(0.0077)
Metz	MSE(std)	0.8133(0.0091)	0.8256(0.0101)	0.3314(0.0111)	0.3099(0.0314)	**0.2864**(0.0048)
	MAE(std)	0.6610(0.0033)	0.6656(0.0042)	0.4319(0.0080)	0.4176(0.0192)	**0.4001**(0.0041)
	R^2(std)	0.1057(0.0100)	0.1032(0.0112)	0.6357(0.0122)	0.6600(0.0345)	**0.6880**(0.0088)
KIBA	MSE(std)	0.5603(0.0058)	0.5624(0.0052)	0.1974(0.0077)	0.2056(0.0025)	**0.1484**(0.0066)
	MAE(std)	0.4050(0.0021)	0.4062(0.0031)	0.2509(0.0052)	0.2833(0.0030)	**0.2128**(0.0040)
	R^2(std)	0.1948(0.0083)	0.1930(0.0079)	0.7186(0.0109)	0.7044(0.0043)	**0.7881**(0.0055)

which shows that in the case of insufficient training data, DeepDTA and GraphDTA are prone to overfitting. Similarly, the effect of AttentionDTA is also affected, but it is relatively stable in comparison. On the KIBA dataset, the trend of declining model performance has moderated, which shows that as the amount of data increases, the model has learned useful features to a certain extent. AttentionDTA yields the best performance on KIBA. But, we can find that AttentionDTA slightly surpasses GraphDTA on KIBA under the setting S2, which implicates that the graph network model precessing on graph structures has better drug feature extraction capability than 1DCNN precessing on SMILES strings.

From table 5, AttentionDTA and DeepDTA show the similar performance on Davis and KIBA datasets, but on Metz dataset, AttentionDTA is better than DeepDTA. In the case of limited protein information input, AttentionDTA has more parameters than DeepDTA, and it is easier to overfit and produce a poor performance in theory. On Davis, AttentionDTA is trained to overfitting after a dozen epochs, while the training of DeepDTA and GraphDTA requires nearly a hundred iterations to gradually converge, and the training time far exceeds that required by AttentionDTA. The performance of AttentionDTA on the Metz and KIBA datasets shows that under the reference of the drug feature, the attention mechanism does help the model extract more effective protein features than these extracted from only 1D-CNN or GCN.

As we all know, the setting S4 is the most challenging. The performance of each model is considerably limited under the setting S4. From the Table 6, we can clearly find that when both drug and protein information is missing, the performance of AttentionDTA is much higher than those of the baselines. Compared with the results under the setting S2 and S3 that also have missing information, AttentionDTA's performance tends to be stable, and the floating of DeepDTA and GraphDTA is larger. It shows that AttentionDTA is very good at finding effective features in the data under the guidance of appropriate attention mechanism, which makes the model highly robust.

3.6 Ablation Study

To explore the impact of attention mechanisms on the DTA prediction task and to find the most suitable attention mechanism with features extracted from 1D-CNNs, we next evaluated seven models based on different attention mechanisms, called protein-side attention, drug-side attention, co attention, additive attention, cross attention, separate attention, and multi-head attention. We divid these attention mechanisms into two types based on the number of objects they act on: one-side attention mechanism and two-side attention mechanism. One-side attention mechanism just extracts the important subsequences on drugs or proteins while two-side attention mechanism simultaneously extracts the important subsequences on drugs and proteins. The detailed computational procedures of these attention mechanisms are described in the supplementary, which can be found on the Computer Society Digital Library at http://doi.ieeecomputersociety.org/10.1109/TCBB.2022.3170365.

Table 7 shows the prediction performance among these models on the KIBA dataset. As it is shown, the models

TABLE 4
Performance Comparison of Our Model and Baselines on Benchmark Datasets Under the Setting S2

Datesets	Metrics	1-NN_drug	DeepDTA	GraphDTA	AttentionDTA
Davis	MSE(std)	1.0419(0.0181)	0.6255(0.0810)	0.6837(0.0267)	**0.3405**(0.0206)
	MAE(std)	0.5528(0.0065)	0.5054(0.0637)	0.4833(0.0462)	**0.3021**(0.0223)
	R^2(std)	-0.1853(0.0201)	-0.0429(0.1049)	0.0921(0.0056)	**0.4906**(0.0264)
Metz	MSE(std)	0.8278(0.0076)	0.5446(0.0065)	0.5468(0.0251)	**0.5175**(0.0232)
	MAE(std)	0.6604(0.0034)	0.5509(0.0013)	0.5518(0.0093)	**0.5480**(0.0179)
	R^2(std)	0.0930(0.0083)	0.4032(0.0072)	0.3914(0.0275)	**0.4355**(0.0125)
KIBA	MSE(std)	0.5337(0.0066)	0.4209(0.0169)	0.3468(0.0089)	**0.3438**(0.0318)
	MAE(std)	0.3994(0.0029)	0.3983(0.0037)	0.3800(0.0059)	**0.3789**(0.0210)
	R^2(std)	0.2321(0.0095)	0.4036(0.0240)	0.4798(0.0174)	**0.4810**(0.0321)

TABLE 5
Performance Comparison of Our Model and Baselines on Benchmark Datasets Under the Setting S3

Datesets	Metrics	1-NN_protein	DeepDTA	GraphDTA	AttentionDTA
Davis	MSE(std)	1.0523(0.0172)	0.3732(0.0132)	0.5457(0.0236)	**0.3625**(0.0132)
	MAE(std)	0.5602(0.0084)	**0.2963**(0.0256)	0.4711(0.0321)	0.3351(0.0248)
	R^2(std)	-0.1735(0.0303)	0.3612(0.0314)	0.2248(0.0312)	**0.3702**(0.0259)
Metz	MSE(std)	0.8356(0.0078)	0.6515(0.0202)	0.6466(0.0226)	**0.5058**(0.0395)
	MAE(std)	0.6594(0.0043)	0.5887(0.0080)	0.5802(0.0085)	**0.5385**(0.0219)
	R^2(std)	0.0892(0.0076)	0.3099(0.0213)	0.3223(0.0237)	**0.4505**(0.0311)
KIBA	MSE(std)	0.5349(0.0062)	0.4147(0.0156)	0.4643(0.0164)	**0.3878**(0.0334)
	MAE(std)	0.4012(0.0031)	0.4241(0.0119)	0.4679(0.0132)	**0.3823**(0.0097)
	R^2(std)	0.2325(0.0084)	0.4052(0.0121)	0.2586(0.0222)	**0.4456**(0.0170)

TABLE 6
Performance Comparison of Our Model and Baselines on Benchmark Datasets Under the Setting S4

Datesets	Metrics	DeepDTA	GraphDTA	AttentionDTA
Davis	MSE(std)	0.8312(0.0832)	0.9546(0.0763)	**0.6058**(0.0581)
	MAE(std)	0.6238(0.0562)	0.6701(0.0501)	**0.4731**(0.0354)
	R^2(std)	-0.2633(0.0312)	-0.2077(0.0398)	**0.0756**(0.0434)
Metz	MSE(std)	0.8590(0.0245)	0.8238(0.0236)	**0.6542**(0.0437)
	MAE(std)	0.7094(0.0083)	0.6970(0.0092)	**0.6308**(0.0184)
	R^2(std)	0.1773(0.0234)	0.1882(0.0259)	**0.2709**(0.0216)
KIBA	MSE(std)	0.6073(0.0335)	0.5809(0.0653)	**0.5317**(0.0635)
	MAE(std)	0.5132(0.0322)	**0.4976**(0.0296)	0.5077(0.0401)
	R^2(std)	0.1034(0.0092)	0.1862(0.0256)	**0.2262**(0.0441)

with attention mechanism are better than the model without attention mechanism, which indicates that it is necessary to establish the correlation between drug features and protein features in DTA inference process. Two-side attention mechanisms are better than one-side attention, which indicates that finding the critical subsequences of protein and the important local substructures of the drug simultaneously is essential in DTA prediction. Moreover, multi-head attention achieves the best experimental results. It is attributed to the fact that multi-head attention mechanism models the relationships between drugs and proteins from multiple feature spaces, which is consistent with the biological phenomenon that there are several non-covalent interaction types between atoms and amino acids (e.g., hydrophobic interactions and hydrogen bonds).

TABLE 7
Ablation Study on KIBA Dataset

Methods	Type	MSE(std)	MAE(std)	R2(std)
No attention	_	0.1981 (0.0054)	0.2556 (0.0037)	0.7133 (0.0042)
Protein-side attention	One side	0.1802 (0.0082)	0.2424 (0.0114)	0.7428 (0.0112)
Drug-side attention	One side	0.1815 (0.0097)	0.2470 (0.0067)	0.7495 (0.0130)
Co attention	Two side	0.1510 (0.0060)	0.2296 (0.0042)	0.7844 (0.0086)
Additive attention	Two side	0.1514 (0.0104)	0.2208 (0.0111)	0.7838 (0.0147)
Cross attention	Two side	0.162 (0.0016)	0.236 (0.0021)	0.756 (0.0027)
Separate attention	Two side	0.164 (0.0017)	0.234 (0.0030)	0.750 (0.0035)
Multi-head-attention	Two side	**0.1489** (0.0056)	**0.2132** (0.0038)	**0.7875** (0.0055)

3.7 Evaluation for Attention Mechanism

The key advantage of our model over baselines is that the attention mechanism not only improves predictive capability, but also has the ability to locate binding sites. To assess the effectiveness of the attention mechanism for localization of binding sites, we test our model on the IC50 dataset provided by Li *et al.* [48]. This dataset provides the IC50 values and the inter-molecular non-covalent interactions between drugs and proteins. There are 5,340 unique pairs in the IC50 dataset.

After feature extraction by CNN in our model, each feature vector is associated with a receptive field on the amino acid sequence. So, a few of consecutive rows in the protein's feature matrix may contain the same binding site information. Moreover, due to the lack of supervised information about binding sites in the training step, our attention mechanism can only try to locate the possible binding sites. Therefore, we calculate the distribution of the probability of binding sites getting higher attention scores. At first, the protein attention scores (i.e., the protein attention vector α_d) is calculated. Second, among the attention scores containing the same binding site information, we set the maximum of them to be the attention score of that binding site and remove the rest of them to avoid reusing the same binding site information. Finally, we rank the attention scores and

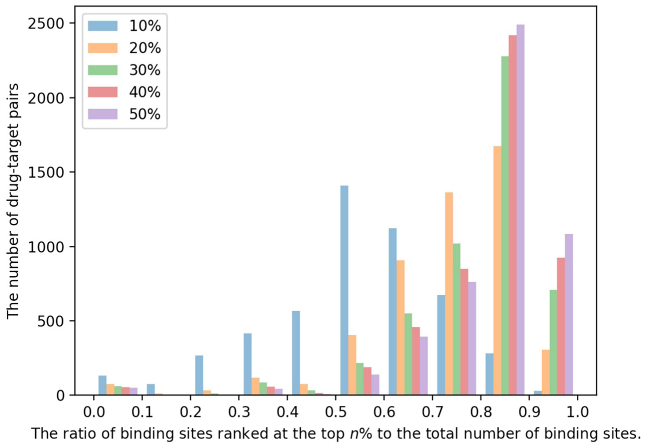


Fig. 2. The histogram of the ratio of the number of binding sites ranked at the top $n\%$ to the total binding sites on proteins.

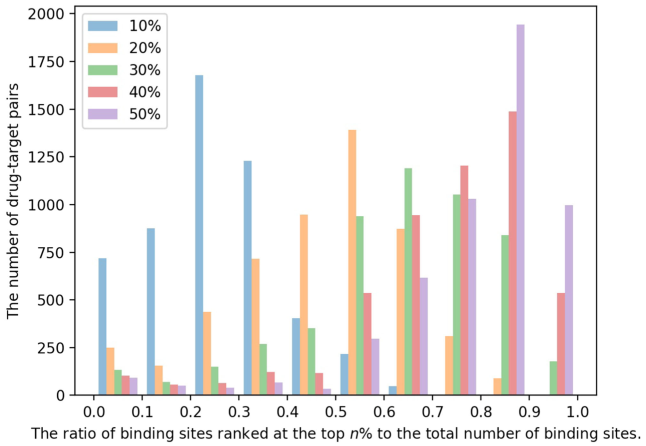


Fig. 3. The histogram of the ratio of the number of binding sites ranked at the top $n\%$ to the total binding sites on drugs.

calculate the ratio of binding sites ranked at the top $n\%$ to the total number of binding sites. For example, if there are 10 binding sites in a protein sequence, and 6 of them rank in the top 10% of attention scores, we define the ratio of the number of binding sites ranked at the top 10% to the total number of binding sites is 0.6. Fig. 2 shows the histogram of the ratio of the number of binding sites ranked at the top $n\%$ to the total binding sites. As we can see in Fig. 2, the ratios of binding sites ranked at the top 10%, 20%, 30%, 40%, and 50%, are in the range of [0.5,0.6], [0.8,0.9], [0.8,0.9], [0.8,0.9], and [0.8,0.9] with the highest statistical frequency, respectively. Precisely, on the whole IC50 dataset, 54.79%, 74.16%, 80.12%, 82.33%, and 83.8% of the binding site attention scores appear in the top 10%, 20%, 30%, 40%, and 50%, respectively.

We performed the same analysis on drugs and the results are shown in Fig. 3. On drugs, the ratios of binding sites ranked at the top 10%, 20%, 30%, 40%, and 50%, are in the range of [0.2,0.3], [0.5,0.6], [0.6,0.7], [0.8,0.9], and [0.8,0.9] with the highest statistical frequency, respectively. On the whole IC50 dataset, 25.5%, 47.94%, 64.74%, 74.15%, and 80.02% of the binding site attention scores appear in the top 10%, 20%, 30%, 40%, and 50%, respectively. Overall, the analysis in proteins and drugs indicates that the binding sites tend to achieve higher attention scores in our model.

3.8 Case Study

To show more intuitively the relationship between attention weights and binding sites, we map the regions with attention weight values calculated from the attention model onto the 3D structure and protein sequence. Here, Fig. 4 shows the attention visualization of the crystal structure of the syk tyrosine kinase domain with Gleevec (PDB ID: 1XBB), and Fig. 5 shows the attention visualization of the crystal structure of serum and glucocorticoid-regulated kinase 1 in complex with compound 2 (PDB ID: 3HDN). On the 3D structure visualization, the regions in protein, which are not binding sites but predicted to be binding sites, are highlighted in red, the regions marked in yellow are true binding sites but not given high attention scores, and the black regions indicate that these regions are true binding sites with high attention scores. On the protein sequence visualization, the attention scores ranging from 0 to 1 are indicated by a color bar with rainbow color gradient. In the

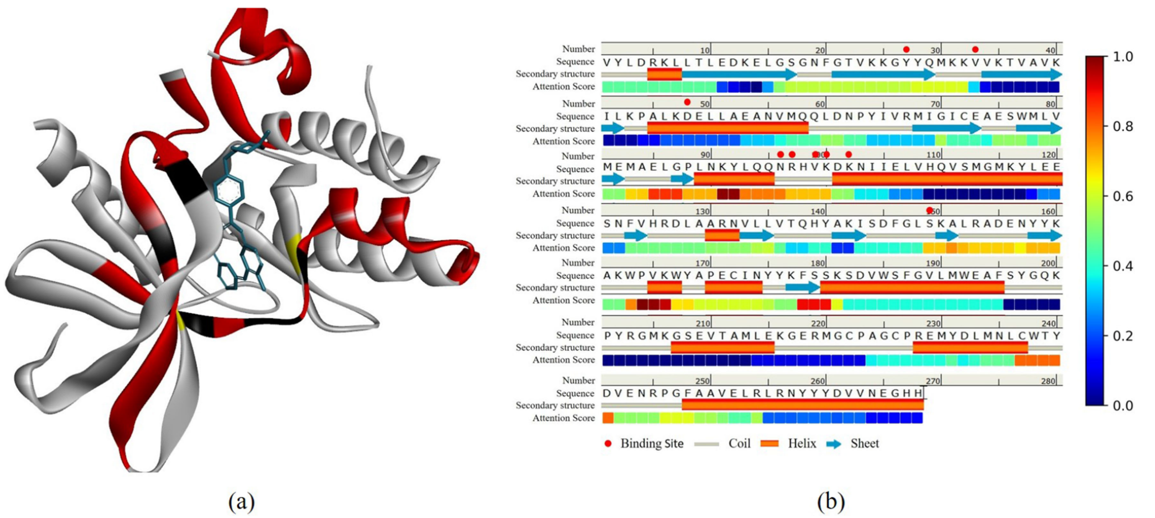
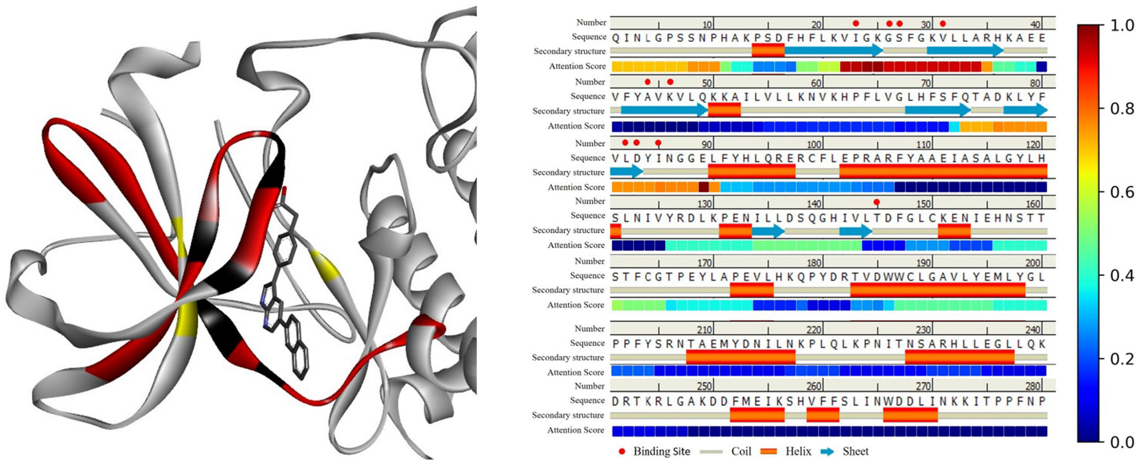


Fig. 4. The attention visualization of 1XBB. (a) The attention visualization on 3D structure. (b)The attention visualization on protein sequence.



(a) (b)

Fig. 5. The attention visualization of 3HDN. (a) The attention visualization on 3D structure. (b)The attention visualization on protein sequence.

case of 1XBB (Fig. 4), eight out of the nine binding sites have a high attention weight (over 0.5). Fig. 5 shows that seven out of the ten binding sites had an attention weight greater than 0.5. From Figs. 4 and 5, the amino acid sequences near the binding sites with high attention scores also have high attention score, which indicate that these amino acids are likely involved in the construction of protein pocket at the binding sites. Thus, our model provides reasonable cues on the binding sites of complex, which is helpful for the visualizing and analyzing the protein molecule model.

4 Conclusion

In this paper, we propose an end-to-end deep learning-based model, AttentionDTA, which uses attention mechanism to find the weight relationships between drug subsequences and protein subsequences to obtain more effective drug and protein representations. In order to get a comprehensive comparison, we set four different data division settings. In general, AttentionDTA has achieved good performance in all four experimental settings. Especially in the most challenging S4 experimental setting, the best prediction results are obtained in predicting novel interactions between new drugs and new proteins. The best performance in the setting S4 shows that our model has a strong ability to extract drug-protein binding features, compared with the independent extraction of drug and protein features in the baselines. The results also show that our model has a more significant improvement on the KIBA dataset, and the size of the KIBA dataset is about four times that of Davis. Therefore, we infer that as the amount of data increases, our model can better learn the representations of drugs and proteins and predict the DTAs. The biggest difference between AttentionDTA and the baselines is the two-side multi-head attention mechanism, which models the non-covalent interactions between atoms and amino acids. We test our model on the IC50 dataset, which provides the binding sites between drugs and proteins. The results show that even if the information of the binding sites is never input during the inference process, our model can still effectively enhance the role of the features at the binding sites. In addition, the attention mechanism provides good visualization for the location of the binding sites on 3D structure and protein sequence, allowing us to analyze the model well.

In recent years, the adaptive processing of graph data has received extensive attention from the deep learning community and has become a mainstream research topic [57]. It is worth noting that the molecular graph structure is also an intuitive and effective way of representing compounds. The use of graph network models to process drugs has been successful in [58], [59], [60]. In the future, we will introduce graph neural networks into our framework to improve the model's ability to predict drug protein affinities.

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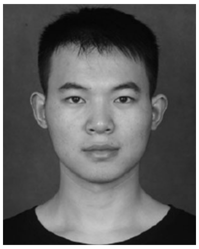
Qichang Zhao is currently working toward the PhD degree with the Hunan Provincial Key Lab on Bioinformatics, School of Computer Science and Engineering, Central South University, Hunan, China. His current research interests include machine learning, deep learning, and bioinformatics.

Guihua Duan received the PhD degree in computer science and technology from Central South University in 2010. She is currently an associate professor with the School of Computer Science and engineering, Central South University. Her research interests include bioinformatics, network security and computer network.

Mengyun Yang received the PhD degree in computer science and technology from Central South University in 2020. He is currently an associate professor with the School of Science, Shaoyang University, Hunan, China. His current research interests include machine learning, deep learning, and bioinformatics.



Zhongjian Cheng is currently working toward the master's degree with Hunan Provincial Key Lab on Bioinformatics, School of Computer Science and Engineering, Central South University, Hunan, China. His research interests include bioinformatics and machine learning.

Yaohang Li received the BS degree from the South China University of Technology in 1997, and the MS and PhD degrees in computer science from Florida State University, Tallahassee, FL, USA, in 2000 and 2003, respectively. He is an associate professor in computer science with Old Dominion University, Norfolk, VA, USA. His research interests are in protein structure modeling, computational biology, bioinformatics, Monte Carlo methods, Big Data algorithms, and parallel and distributive computing.

Jianxin Wang (Senior Member, IEEE) received the BEng and MEng degrees in computer engineering from Central South University, China, in 1992 and 1996, respectively, and the PhD degree in computer science from Central South University, China, in 2001. He is the dean and a professor with the School of Computer Science and Engineering, Central South University, Changsha, Hunan, China. His current research interests include algorithm analysis and optimization, parameraized algorithm, bioinformatics, and computer network.

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